

Name: DARISY SREETEJRoll No: EE25BTECH11018Instructions

1. The exam duration is 90 minutes, starting at 12:10. Ensure that you submit both the question paper and the answer book by 13:40. Please be aware that **late submissions** may incur **penalties**.
2. Read all questions carefully. Misinterpretations will **not** be considered. It is essential to write your solutions clearly and legibly in the answer book. In a previous quiz, a few students submitted answers that were not clearly written. Going forward, such responses will **not** be evaluated.
3. Any form of unfair practice will result in an **F grade** and will be reported to the disciplinary committee.

1. (5 points) Let $v(t)$ denote a triangular wave of frequency 0.25 Hz with zero average and an RMS value of $10\sqrt{3}$ V. Compute the average value and RMS value of the signal $v^2(t)$. **Note:** You are free to choose a triangular wave of your choice, as long as it satisfies the given conditions. Clearly plot the waveform you have chosen, mark the important points on the waveform, although it may or may not affect the calculations.
2. (5 points) Consider the circuit shown in Fig. 1(a). The waveform of the voltage signal applied to the inductor is shown in Fig. 1(b). Assume that the initial current through the inductor is 0.

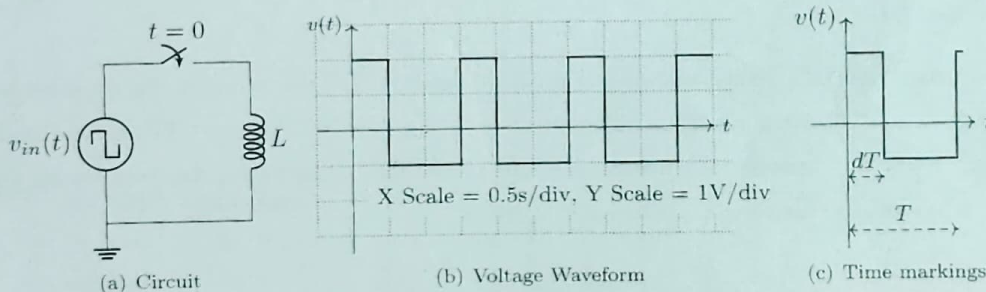


Fig. 1: For Question 2

- (a) (2 points) Plot the current through the inductor for $t > 0$ (for atleast 2 cycles). Clearly indicate the points where the slope of the current changes and the peak value of the current.
- (b) (1 point) Compute the average value of current for the first cycle of the voltage waveform.
- (c) (2 points) It is desired to adjust the transition between the positive and negative portions of the voltage waveform such that the average current through the inductor is zero. Let d denote the fraction of each cycle during which the voltage is positive. Compute the value of d . (See Fig. 1(c) for reference). The magnitudes of the positive and negative voltages remain unchanged. You may assume that ensuring zero average current during the first cycle is sufficient to guarantee zero average current in all subsequent cycles.

3. (5 points) The steady-state voltage and current through a capacitor are shown in Fig. 2. Note that the voltage and current waveforms are not explicitly labeled and you must deduce it based on your understanding. For the following

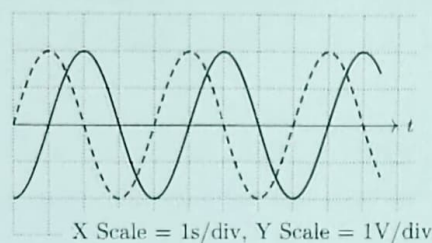
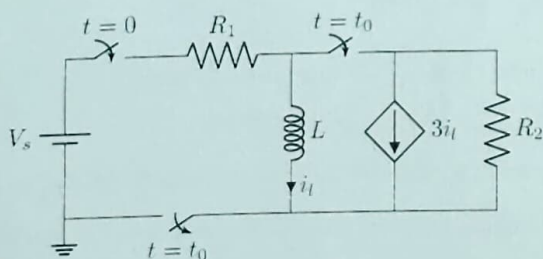


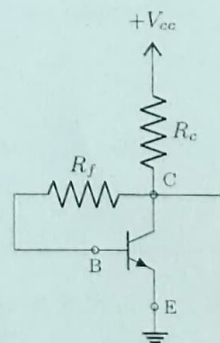
Fig. 2: Voltage and Current Waveforms for Question 3

questions, pick a time reference of your choice and clearly indicate it on the plot provided in the question paper.

- (1 point) Determine the value of the capacitance C .
 - (1 point) Determine the phasor representations of the voltage and current waveforms. Clearly indicate the time reference you have chosen.
 - (2 points) Advance the time reference you chose earlier by $T/4$. Compute the phasor representations of the voltage and current waveforms with respect to this new reference. Clearly mark both the original and the new time references on the plot in the question paper.
 - (1 point) Determine the instantaneous reactive power $q(t)$ associated with the capacitor.
4. (5 points) Consider the circuit shown in Fig. 3(a). Compute the current $i(t)$ through the inductor for $t > 0$. Assume that the initial current through the inductor is zero. Assume that t_0 is large enough for the circuit to have reached steady state before t_0 .
5. (5 points) Consider the BJT circuit shown in Fig. 3(b). Assume that the voltages and resistors in the circuit are chosen such that the BJT is operating as an amplifier (i.e., $V_{BE} > 0.7V$; Assume $0.7V$ as the threshold voltage for this problem). Prove or disprove the following statement: The BJT cannot be driven into saturation (ON-state) by varying the values of any passive components/DC supplies.



(a) for Question 4



(b) for Question 5

Fig. 3: Circuits for questions 4 and 5